An energy transport over a distance of several micrometers cannot rely on electron diffusion because for energies up to 500 eV the penetration depth of an electron beam in water is below 30 nm [288], and the diffusion length for ambipolar diffusion of electrons and ions is even smaller. Electrons are thus largely confined to the region in which they have been produced, which explains the small lateral dimension of the strings that grow in length with the moving breakdown wave. Other than electrons, 500-eV photons in the water-window have a penetration range comparable to the extent of the halo [286]. Such energetic photons may arise from bremsstrahlung emitted by hot electrons in the strings. When a very high ionization level is reached, the ongoing laser energy deposition quickly raises the average electron temperature, and the photon energy of bremsstrahlung emitted from the electrons increases [289]. Initially merely UV bremsstrahlung is emitted, which is re-absorbed within a very short distance because the penetration depth for linear absorption in water is below 300 nm for photon energies between 7.7 eV and 128 eV [286]. This way, incident laser energy remains trapped in the strings, while electrons gain further energy. Finally, electrons are sufficiently hot for the generation of X-ray bremsstrahlung that can leave the region of initial energy deposition and produce plasma elsewhere. The penetration depth of the soft X-rays increases to 10 µm when the photon energy grows to 500 eV (λ = 2.5 nm) and further to 30 µm at 2.5 keV (λ = 0.5 nm). Such penetration depths are comparable to the extent of the luminescent halo beyond the laser cone angle. The possibility of soft X-ray emission in ns breakdown in water is corroborated by earlier experiments in which photons in the 500-eV region were emitted from ns-laser-produced plasmas on a thin water jet passing through a vacuum chamber [238,239].

Why can X-ray photons escape the high-density plasma strings? For extreme UV radiation and soft X-rays, inverse bremsstrahlung absorption is the dominant absorption mechanism both in native water (Fig. 14) and in plasma [37,290]. Due to the increasing attenuation length with decreasing wavelength (or increasing photon energy, respectively), X-rays are only weakly reabsorbed in the plasma. Even for $\rho_c = 6.68 \times 10^{22}$ cm^{-3} corresponding to full ionization of the $1b_1$ valence band orbital, the frequency of X-rays with wavelengths shorter than 129 nm exceeds the plasma frequency. Therefore, no 'plasma mirror' evolves and X-rays can readily penetrate into the surrounding lower-density plasma or water.

The elongation of the halo in the propagation direction of the laser beam seen in Figs. 7(c) and (e) indicates that X-ray emission occurs preferentially in string direction. A possible explanation for the emission anisotropy is a field gradient from string tip towards stem favoring bremsstrahlung emission along the string direction, which evolves during string growth. Simulations of streamer growth in gases showed that the different motility of electrons and ions results in a charge separation [265,275], and we

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assume that the same applies for the strings observed in our experiments. Electrons and ions are simultaneously generated at the string tips, and here the local charge is neutral. During string growth, electrons in its stem move towards the periphery because equal charges repel each other, while the heavier positive ions remain in the center. This way, a potential difference evolves along the string axis from the neutral tip towards the positive core of the stem. The resulting electric field may accelerate electrons preferentially towards the stem, i.e. towards the laser beam waist, which would promote the generation of energetic bremsstrahlung in this direction.

X-ray absorption in the halo region creates a cascade of secondary low-energy electrons [160]. When the energy of these electrons drops below the ionization threshold of water, they produce no more electrons by impact ionization and are thermalized by collisions, hydration, and recombination. Bremsstrahlung emitted by electrons with energies of a few eV has an emission peak in the UV but the long-wavelength tail lies in the visible region of the optical spectrum [290]. This tail is transmitted by water, can leave the breakdown region and is visible on the photographs. Therefore, the plasma luminescence has a similar bluish appearance everywhere in the diffuse halo region, regardless of spatial variations of the initial kinetic energy distribution before thermalization. The radiation energy leaving the water cell amounts to less than 0.1 % of the absorbed laser energy [114] but a much larger fraction of the absorbed laser energy must be involved in radiative energy transport within the plasma region itself, where electron temperatures are higher and the conversion into bremsstrahlung is more efficient [291,292].

Due to the limited spectral range of our digital camera, an experimental exploration of the wavelength dependence of string formation was not possible and remains a task for future research. Since strings arise from an intrinsic spatial instability of AI and because AI becomes ever stronger with increasing wavelength, the phenomenon is expected to be a prominent feature also in ns IR breakdown but weak in UV plasmas. Moreover, the critical electron density is larger for shorter wavelengths, which promotes 'regular" nonlinear absorption and delays the onset of the absorption/reflection runaway at $\rho_c \approx \rho_{crit}$. Both factors together will tend to level out small-scale inhomogeneities at 355 nm compared to 532 nm. Nevertheless, UV plasmas produced at large *NA* inflate beyond the pump laser cone angle as seen in Fig. 7(b).

Let us now look at the plasma growth and radiative energy transport for ns plasmas produced at moderate numerical aperture, which is shown in Fig. 8 for $NA = 0.3$ and $\lambda_L = 355$ nm. Here we do not see a region of primary energy deposition surrounded by a diffuse halo but rather a sharply delineated,

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mottled region extending from the beam waist in upstream direction, with a bright spot at the upper end. The bright spot is particularly pronounced and large at E_L = 575 µJ, where it has a blurred, reddish periphery. The upstream movement of the breakdown wave is here faster than at large *NA*s (the average speed is 27 km/s for E_L = 575 µJ), and the rate of avalanche ionization is smaller than for breakdown at 532 nm, *NA* = 0.8 discussed above. Therefore, the average plasma energy density reached during the initial upstream movement of the breakdown wave is smaller. Nevertheless, a very high ionization level may be reached in the hot spot, where the energy is deposited during the second half of the laser pulse. For E_L = 575 µJ, we deduced a peak plasma density of 260 kJ/cm^3 in the hot spot. This value is almost as large as the estimate of 400 kJ/cm^3 for the high-density strings observed at large *NA*. How can such large energy densities be produced in a relatively large volume? The increase of critical electron density with decreasing wavelength [Eq. (14)] already implies the possibility of reaching large plasma energy densities. If additionally ρ_c exceeds ρ_{crit}, the resulting drop in absorption depth can lead to an even more efficient heating of this plasma part by the ongoing laser irradiation in the second half of the laser pulse. The location, where ρ_{crit} is first exceeded, depends on a balance between geometrical focusing of the pump laser beam and energy depletion by its absorption within already existing plasma. This location will lie somewhere in the upstream part of the plasma. From there, a front with increased absorption/reflection will move towards the incoming laser beam following the original breakdown wave. The reflected part of the incident laser energy is absorbed within the already existing plasma, close to the reflecting layer, and bremsstrahlung emitted by the highly absorbing region is largely re-absorbed within the cone angle of the laser beam too. This way, a large hot spot is formed.

Plasma radiation promotes plasma growth in axial direction: the plasma length is 50 µm already at the BPL threshold for *NA* = 0.3, which is much larger than the Rayleigh length of the focal region (18.5 µm). Present findings on plasma prolongation and inflation are consistent with previous observations on IR ns breakdown at moderate *NA* [108,293], which suggested that plasma radiation lowers the BPL threshold in the vicinity of the plasma and promotes the formation of large plasmas already at and close to the breakdown threshold.

The *NA* dependence of plasma energy density in ns breakdown differs from that in fs breakdown, where U_{avg} drops dramatically with decreasing NA. For slm ns pulses, U_{avg} remains approximately constant between *NA* = 0.8 and *NA* = 0.3 (Table 3). This similarity relies on two counteracting trends: 1. The optical breakdown wave moves faster at moderate *NA*s than for very tight focusing, which reduces U_{avg}. 2. On the other hand, plasma radiation is at moderate *NA*s re-absorbed within the region of primary

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